CHEM 5260 Midterm Project

Background

In class we considered a system involving a single energy state, coupled to an infinite continuum of states. For this system the Hamiltonian was:

$$\hat{H} = \hat{H}_0 + \hat{V} \tag{1}$$

$$\hat{H}_0 = E_1 \left|1\right\rangle \left\langle 1\right| + \sum_l E_l \left|l\right\rangle \left\langle l\right| \tag{2}$$

$$\hat{V} = \sum_l \Big(V_{1,l} \left|1\right\rangle \left\langle l\right| + V_{l,1} \left|l\right\rangle \left\langle 1\right| \Big) \tag{3}$$

Through various transformations we were able to analytically solve the Schrödinger equation for this Hamiltonian, obtaining the result:

$$\left|C_1(t)\right|^2 = e^{-\Gamma_1 t/\hbar} = e^{-k_1 t} \tag{4}$$

with

$$k_1 \equiv \frac{\Gamma_1}{\hbar} = \frac{2\pi}{\hbar} \overline{\left|V_{1,L}\right|^2} \rho_L \tag{5}$$

While this result is incredibly useful, for practical applications of computational chemistry, analytical solutions are almost always impossible to find, and thus, it is vital to be able to solve this same problem and similar problems using numerical methods.

We begin the numerical method by first considering the Hamiltonian. The form of the Hamiltonian is the same as before, but with a few necessary changes:

$$\hat{H} = E_1 \left|1\right\rangle \left\langle 1\right| + \sum_{k=0}^{N} E_k \left|k\right\rangle \left\langle k\right| + \sum_{k=0}^{N} \Big(V_{1k} \left|1\right\rangle \left\langle k\right| + V_{k1} \left|k\right\rangle \left\langle 1\right| \Big) \tag{6}$$

Firstly, we are no longer able to consider an infinite number of states reaching to $\pm\infty$ and we truncated to some N number of states, in a range ΔE. We will choose to have our ΔE extent evenly above and below our ground state energy which we shall choose to be 0 (an arbitrary choice). We construct our manifold by taking $\varepsilon = \Delta E/N$ with $\rho = 1/\varepsilon = N/\Delta E$. The Hamiltonian mentioned above is the case we considered in class where a single state is coupled to a continuum of states $\left|k\right\rangle$, and we shall call this **System A**.

We shall also consider another system, which we shall call **System B**, in which a state $|0\rangle$ is coupled to a state $|1\rangle$ and state $|1\rangle$ is coupled to a continuum of states $|k\rangle$. The Hamiltonian describing this system is below and follows the same choices made above with respect to ΔE and N:

$$\hat{H} = E_0 \left|0\right\rangle \left\langle 0\right| + E_1 \left|1\right\rangle \left\langle 1\right| + V_{01} \left|0\right\rangle \left\langle 1\right| + V_{10} \left|1\right\rangle \left\langle 0\right| + \sum_{k=0}^{N} E_k \left|k\right\rangle \left\langle k\right| + \sum_{k=0}^{N} \Big(V_{1k} \left|1\right\rangle \left\langle k\right| + V_{k1} \left|k\right\rangle \left\langle 1\right| \Big) \tag{7}$$

The numerical approach for this Hamiltonian is very simple. Since the Hamiltonian is a finite matrix, its eigenvalues and eigenfunctions can be easily calculated, corresponding to the energies and eigenstates of the systems respectively. From here the time evolution operator can be applied to our chosen state (being state $|1\rangle$ for System A and state $|0\rangle$ for System B) to each state, since we have already found the energy of each eigenstate. Then we simply take the square of the inner product of our initial/ground state and our time evolved state:

$$c_j = \langle j \rangle \, \psi(0) \tag{8}$$

$$|\psi(t)\rangle = \sum_j c_j \, e^{-iE_j t} \, |j\rangle \tag{9}$$

$$P_1(t) = \left| \sum_j c_j \, e^{-iE_j t} \, \langle 1 \rangle \, j \right|^2 \tag{10}$$

System A Results

First we examine System A.

We want to satisfy $\Delta E \gg V_{1,k} \gg \rho$.

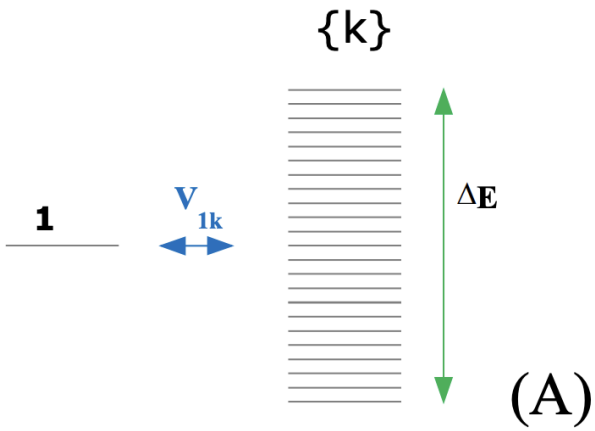


Figure 1: System A energy level diagram.

Let's see how many states we need in our continuum in order to gain accurate results. To do this we vary N from 50 to 300 with steps of 50. Lets begin by seeing how large of a continuum we need in order to approach the analytical result.

Figure A.1 $N = 50$, $V = 0.5$

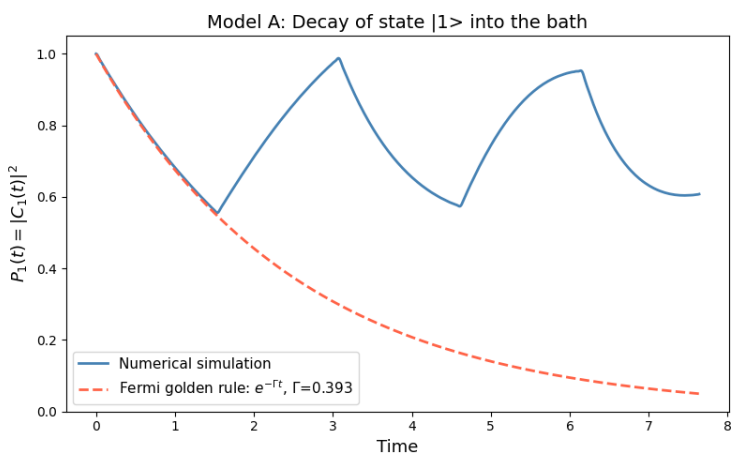


Figure A.2 $N = 100$, $V = 0.5$

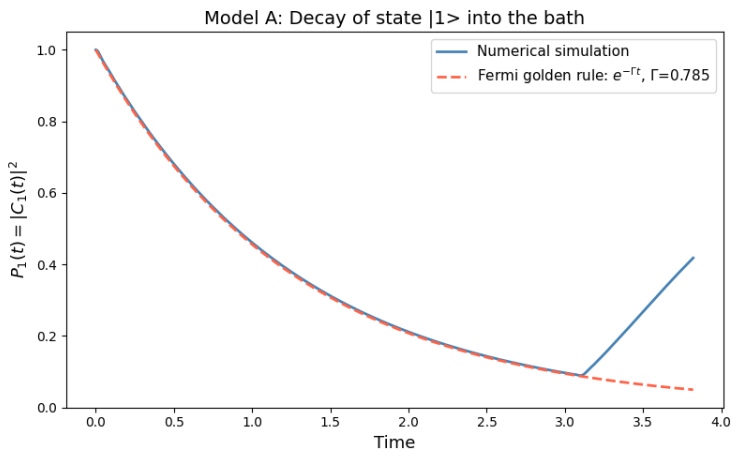


Figure A.3 $N = 150$, $V = 0.5$

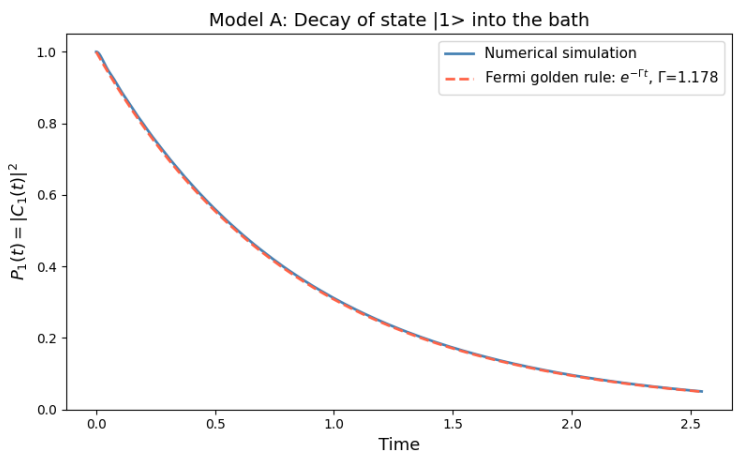
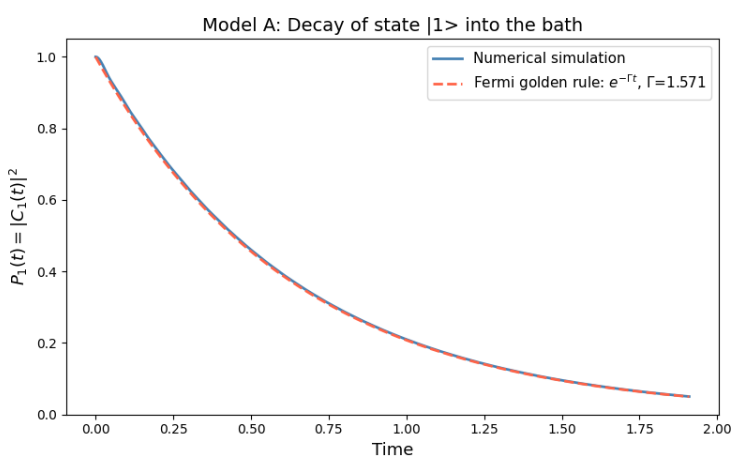


Figure A.4 $N = 200$, $V = 0.5$



From $N = 100$ to 150 we achieved a sufficiently large number of states to match the

analytical results. This corresponds to $\rho = 1$ if we take N to be 200. I will use $N = 200$ for the rest of the calculations.

Recalling that we want to stay around $\Delta E \gg V_{1,k} \gg \varepsilon$ (which is true so long as ΔE is roughly equal to or greater than 200), let's vary our ΔE (and consequently our ρ) going from $\Delta E = 100$ to $\Delta E = 400$ in steps of 100.

Figure A.5 $\Delta E = 100$, $\rho = 2$, $V = 0.5$

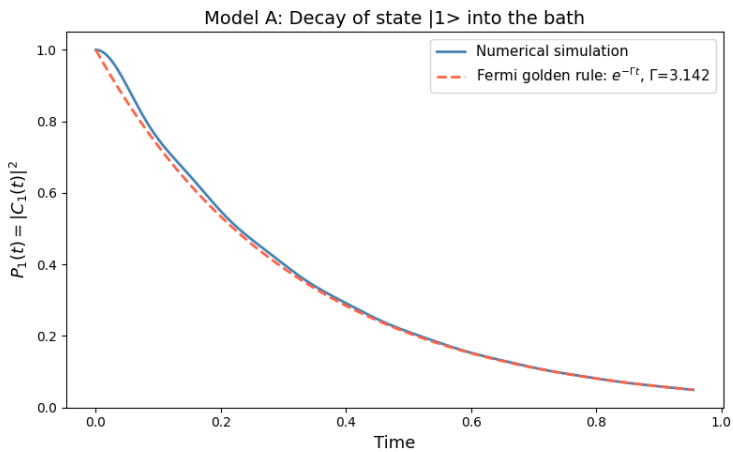


Figure A.6 $\Delta E = 200$, $\rho = 1$, $V = 0.5$

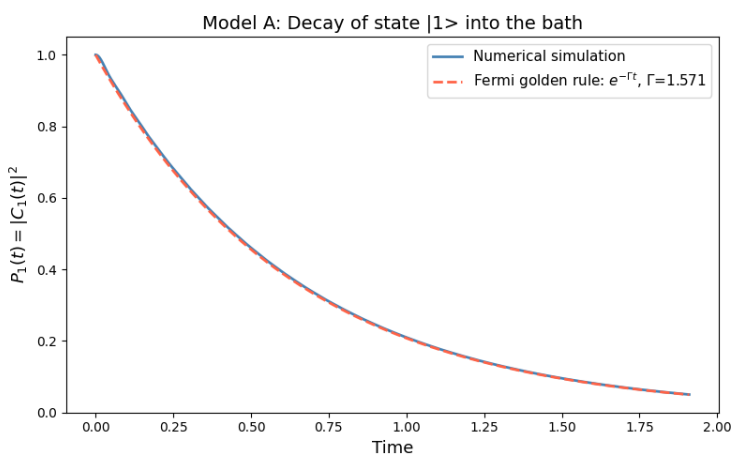


Figure A.7 $\Delta E = 300$, $\rho = 2/3$, $V = 0.5$

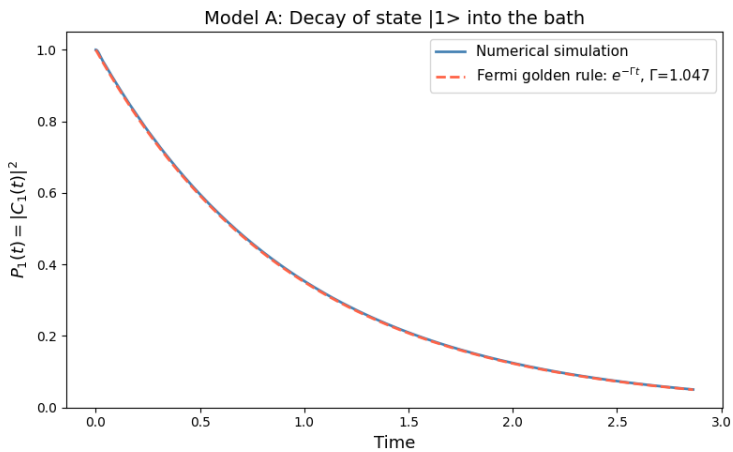
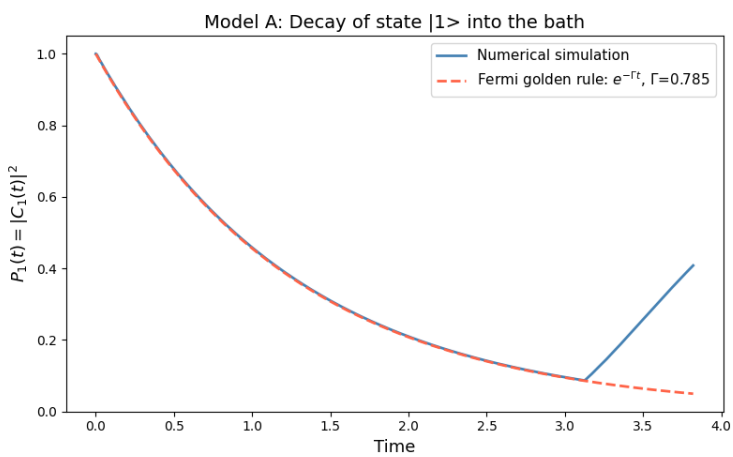


Figure A.8 $\Delta E = 400$, $\rho = 1/2$

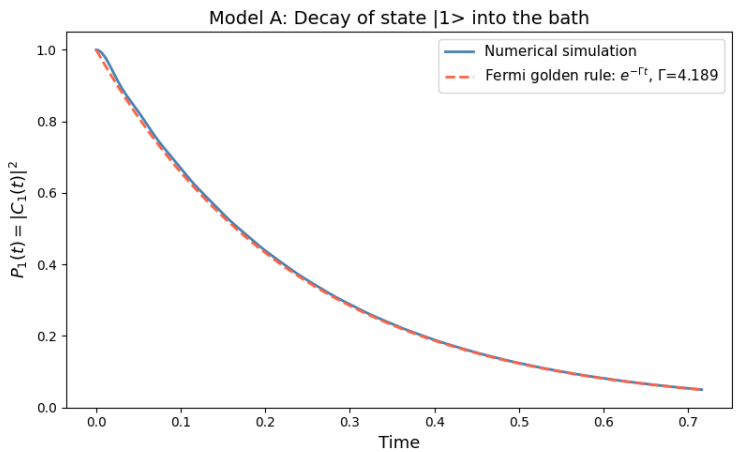


We see that moving from $\Delta E = 300$ to $\Delta E = 400$ the numerical model fails as we reach what is essentially a periodic state of the system that arises from the fact that we have a finite number of basis functions. We see that as the number of basis states becomes too sparse in our continuum, i.e. as ε becomes too large, the model breaks down as we begin to see its periodic image.

We find that $N = 200$ and $\Delta E = 200$ will be good for choosing our constants — as was suggested in the outline, however we have been using $V = 0.5$ which does not satisfy the given rule, $E \gg V_{1,k} \gg \varepsilon$, despite the fact that the models match in this range.

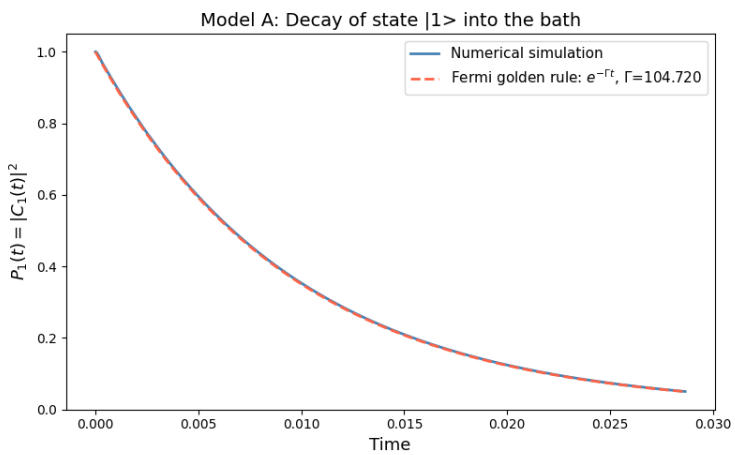
In order to make the models match we can consider another case where we raise ΔE to something like 300 which then allows us to raise V to $V = 1$, both matching the analytical model and satisfying $E \gg V_{1,k} \gg \varepsilon$ (see Fig. A.9 below).

Figure A.9 $\Delta E = 300$, $\rho = 2/3$, $V = 1$



We can also see that the relation $E \gg V_{1,k} \gg \varepsilon$ holds true for more extreme cases; for example we have model agreement for $\Delta E = 30\,000$ and $V = 50$ (i.e. strong coupling to a massive continuum of states, even though $N = 200$).

Figure A.10 $\Delta E = 30\,000$, $\rho = 2/3$, $V = 50$



Discussion

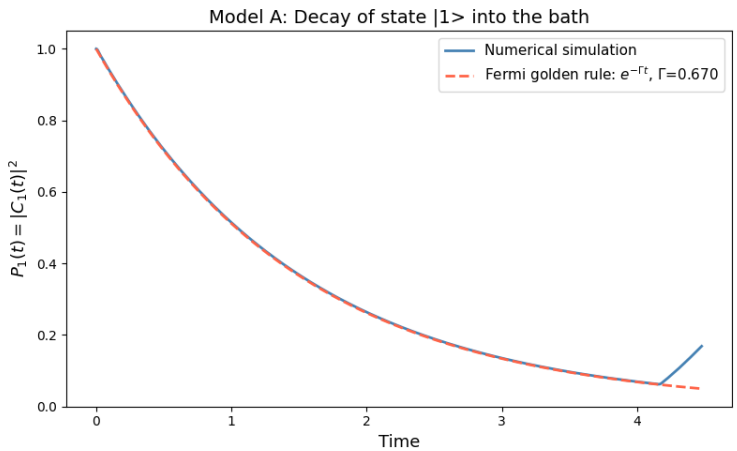
I refer to us seeing the periodic image of the solution, but let's examine what is really going on here. This periodic image is what are called Poincaré recurrences. Unlike with the analytical results, the numerical system necessarily consists of a finite number of basis functions. This finiteness introduces a non-physical phenomenon of returning back to our ground state at periodic times. We see that these images are brought into view for large ε values.

The other deviation we see from the analytical solution occurs when we make ΔE small. We do not see any periodicity, but rather the curve begins to deform away from that of exponential decay. This is simply because the analytical result is derived for the case of an infinite continuum of states, and as we make ΔE small we deviate from this result. In other words, in order for our numerical result to match the analytical result we need a sufficiently large ΔE to approximate an infinite continuum.

Let's now look at what range of V the numerical method holds for, for $\Delta E = 300$ and $N = 200$.

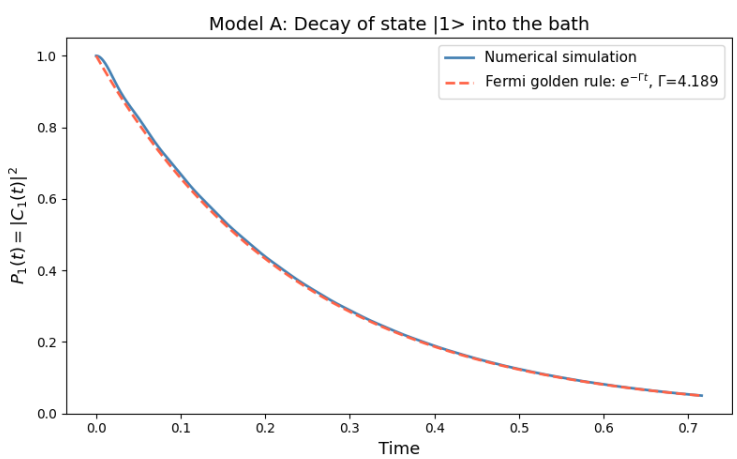
We see the model has a lower limit of $V = 0.5$ and then subsequently breaks around $V = 0.4$.

Figure A.11 $\Delta E = 300$, $\rho = 2/3$, $V = 0.4$



We can then approach the upper limit, and we see that the numerical method works well until around $V = 1$ where we get a noticeable deviation from the exponential shape, although the point at which the model fails is somewhat less clear here as it is a matter of the curve slowly deviating from exponential rather than seeing the periodic image, so we say that the model holds very well until around 0.8 or so.

Figure A.12 $\Delta E = 300$, $\rho = 2/3$, $V = 1$



We can do the same but for $\Delta E = 400$, and here we see the lower limit is at $V = 0.6$ and the upper limit is at around $V = 1.2$.

Figure A.13 $\Delta E = 400$, $\rho = 1/2$, $V = 0.5$

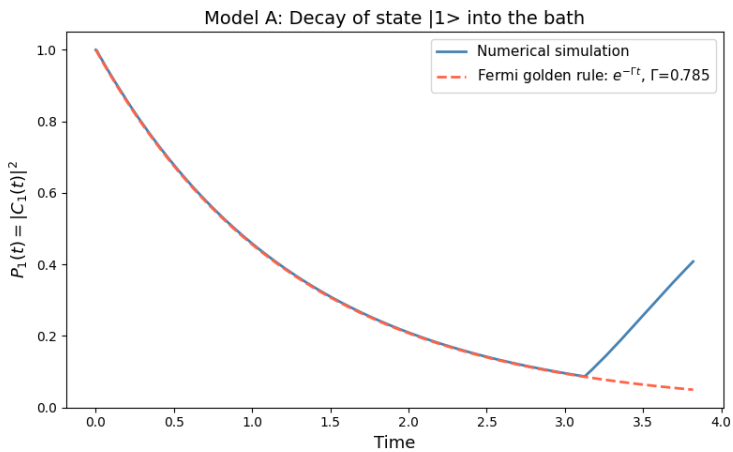
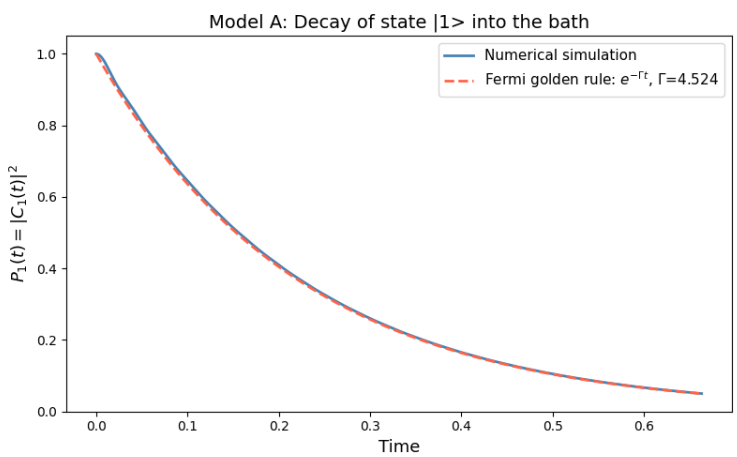


Figure A.14 $\Delta E = 300$, $\rho = 1/2$, $V = 1.2$



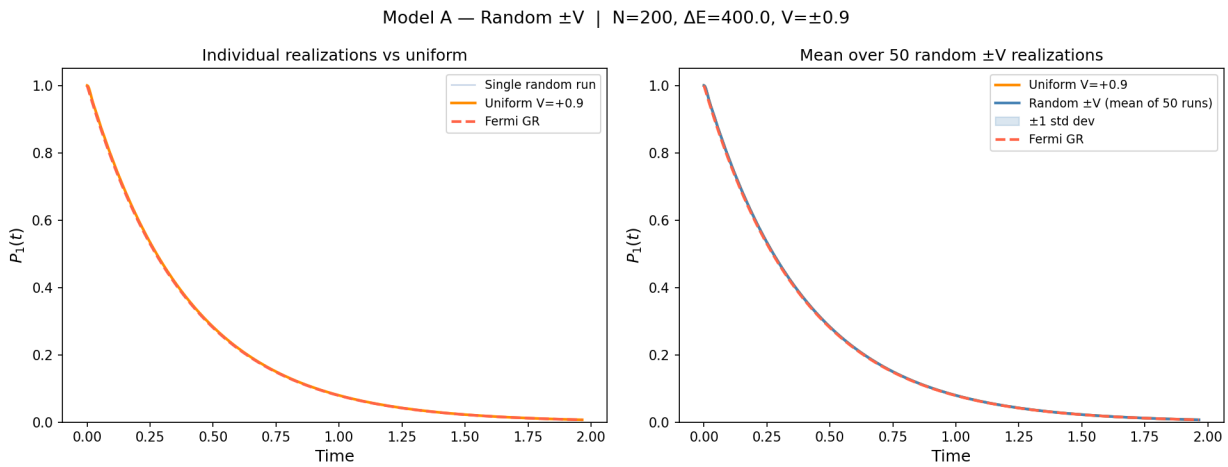
Discussion

We see a similar effect on the appearance of the Poincaré recurrences and on the deviation from exponential decay that we saw before, and we may have guessed this from the relation we were given, $\Delta E \gg V_{1k} \gg \varepsilon$. Essentially, when the coupling is too small it is comparable in size to the line width spacing and our state begins to "see" more of the discrete states rather than a continuum. So just like before this is akin to too large of an ε and thus we break the wide-band approximation.

Similarly, on the other hand, when the coupling is too strong, our system becomes more and more sensitive to the size of our ε. In other words if V is too large, our state can "see" that our continuum is not infinite.

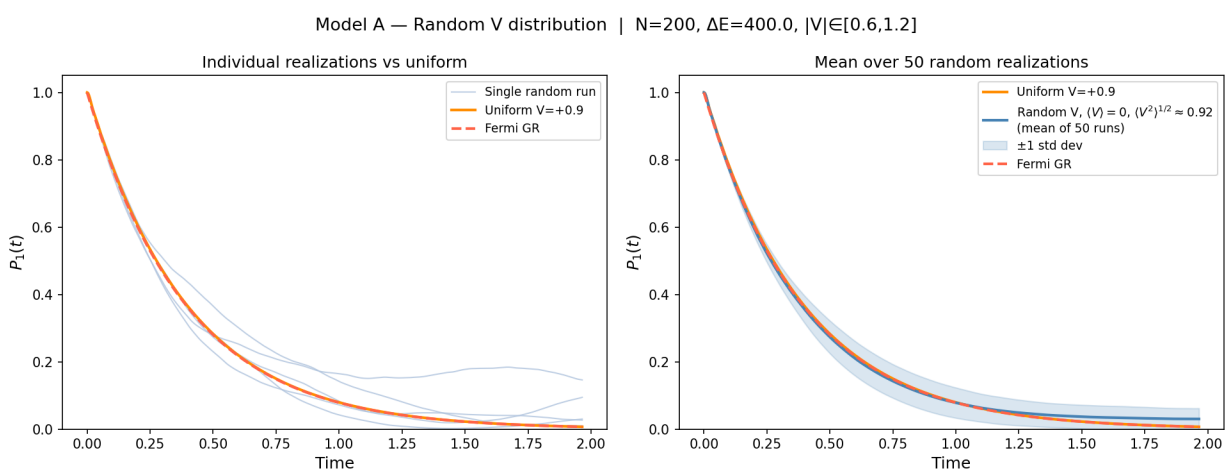
Now we are to study what happens when we take random signs for constant coupling values (constant in magnitude). Let's pick $V = 0.9$ as this is the middle of our acceptable range for the $\Delta E = 400$ case. We display a single run example on the left and the result within one standard deviation of 50 runs on the right, although we see this is not necessary as decay slopes do not change from the normal constant and positive V value runs.

Figure A.15 $\Delta E = 300$, V is randomly assigned to be $+0.9$ or -0.9



We are also to study what happens if we vary the $V_{1,k}$ values such that $\langle V \rangle = 0$, $\langle V^2 \rangle^{1/2} \gg \varepsilon$. Since we have regions of values of V for which the model fails we take the values bounded between -1.2 and -0.6 or $+0.6$ and $+1.2$ (to keep magnitude away from 0). The calculations can be seen below for which the RMS $= 0.917$. We sample a few individual runs shown on the right, and then we take a collection of 50 runs and display one standard deviation.

Figure A.16 $\Delta E = 300$, V is varied over distribution with RMS $= 0.917$



Discussion

For the case in which we randomly take $\pm|V|$ we see no change in the results compared to when we simply took $+|V|$. This is actually a very insightful confirmation of the Fermi Golden Rule which states $\Gamma = 2\pi\rho V^2$; in this we see V is squared and that any sign change cannot affect the final Γ.

For the case where we vary our V around a distribution we see what we would expect: sometimes coupling is stronger and decay occurs more rapidly, and other times it is weaker

and occurs more slowly, but if we average them we obtain the standard result, with increasingly fast or increasingly slow decay being more and more unlikely the more we deviate from the original system.

System B Results

We now consider System B, a system containing two single states in which state $|0\rangle$ is coupled to state $|1\rangle$ and state $|1\rangle$ is coupled to state $|0\rangle$ and a continuum of states $|k\rangle$.

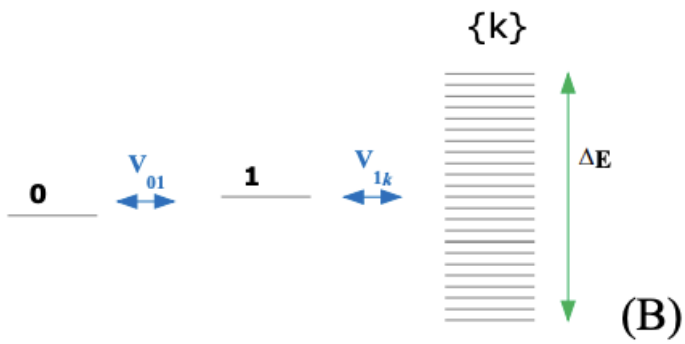


Figure 2: System B energy level diagram.

In the previous set of trials for System A we validated the claim in the outline that $N = 200$ is a good choice. We will keep this value the same but let us examine a range of values for ΔE and our V's. We can compare our results to the known analytical result below for reference:

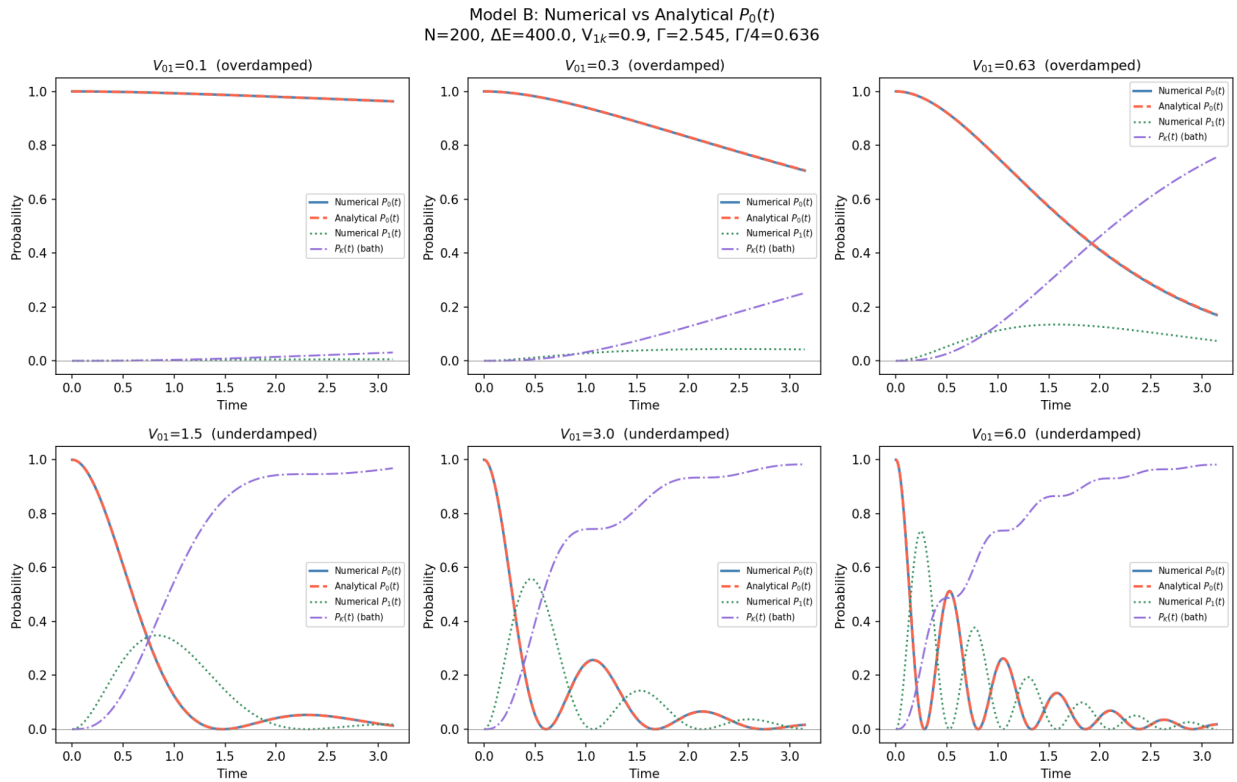
$$P_0(t) = e^{-\Gamma t/2}\left[\cos(\Omega t) + \frac{\Gamma}{4\Omega}\sin(\Omega t)\right]^2 \tag{11}$$

with:

$$\begin{aligned} \Gamma &= \text{decay rate of } |1\rangle \text{ into the continuum} \\ V_{01} &= \text{coupling between } |0\rangle \text{ and } |1\rangle \\ \Omega &= \sqrt{V_{01}^2 - (\Gamma/4)^2} \end{aligned} \tag{12}$$

Let's first look at taking a constant $V_{1,k}$ and varying V_{01}.

Figure B.1 $\Delta E = 400$, $N = 200$, $V_{1,k} = 0.9$, V_{01} varied



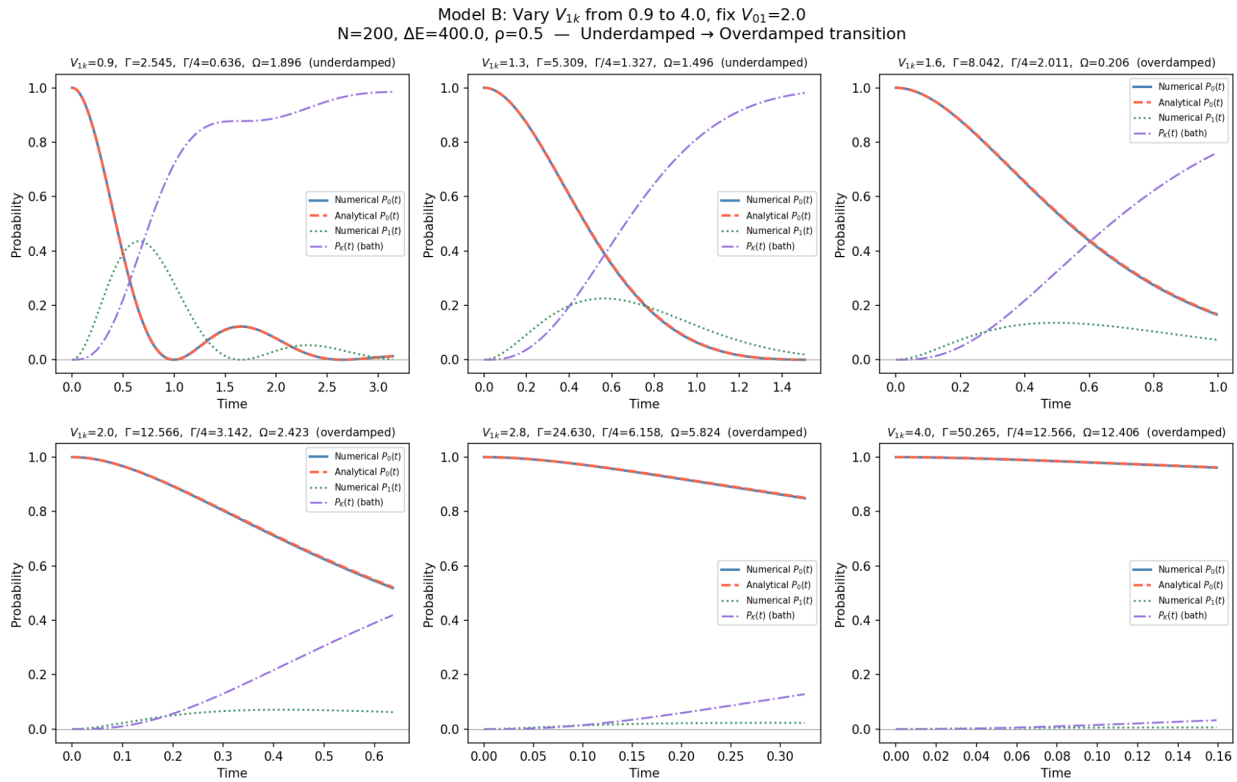
Discussion

Now looking at System B and varying V_{01} we see what we might expect, being that as the coupling between $|1\rangle$ and $|0\rangle$ increases the decay rate into the continuum increases as transfer to $|1\rangle$ is facilitated and as we know we must first enter state $|1\rangle$ before we enter $|k\rangle$. We can think of it like this: due to the large number of states in $|k\rangle$ the system eventually moves from any singular state to a state in $|k\rangle$ as outlined by statistical mechanics, so if states move more easily from the initial state $|0\rangle$ to $|1\rangle$ they increase this rate ($|0\rangle \rightarrow |1\rangle$) which thus increases Γ.

We should also note that there is a point at which we move from the single decay seen in the overdamped case, to where we gain oscillatory behavior in the underdamped case. Thinking in terms of the analytical solution we are reaching the point $\Gamma/4 > V_{01}$ (overdamped) and $\Gamma/4 < V_{01}$ (underdamped) at which the solution to the differential equation changes form to include this oscillatory behavior. More intuitively/physically, what is occurring is that in the overdamped case the V_{01} is sufficiently weak that decay happens quickly from the $|1\rangle$ state as there is not enough time in state $|1\rangle$ to allow for return to state $|0\rangle$. However for the underdamped case, we are increasing V_{01} to the point $\Gamma/4 < V_{01}$ at which the coupling is sufficiently strong and some states will oscillate back and forth between $|0\rangle$ and $|1\rangle$ with a slow decay into $|k\rangle$ at the same time.

We now look at the alternate case of holding V_{01} constant and varying $V_{1,k}$.

Figure B.2 $\Delta E = 400$, $N = 200$, $V_{01} = 2$, $V_{1,k}$ varied (Overdamped)

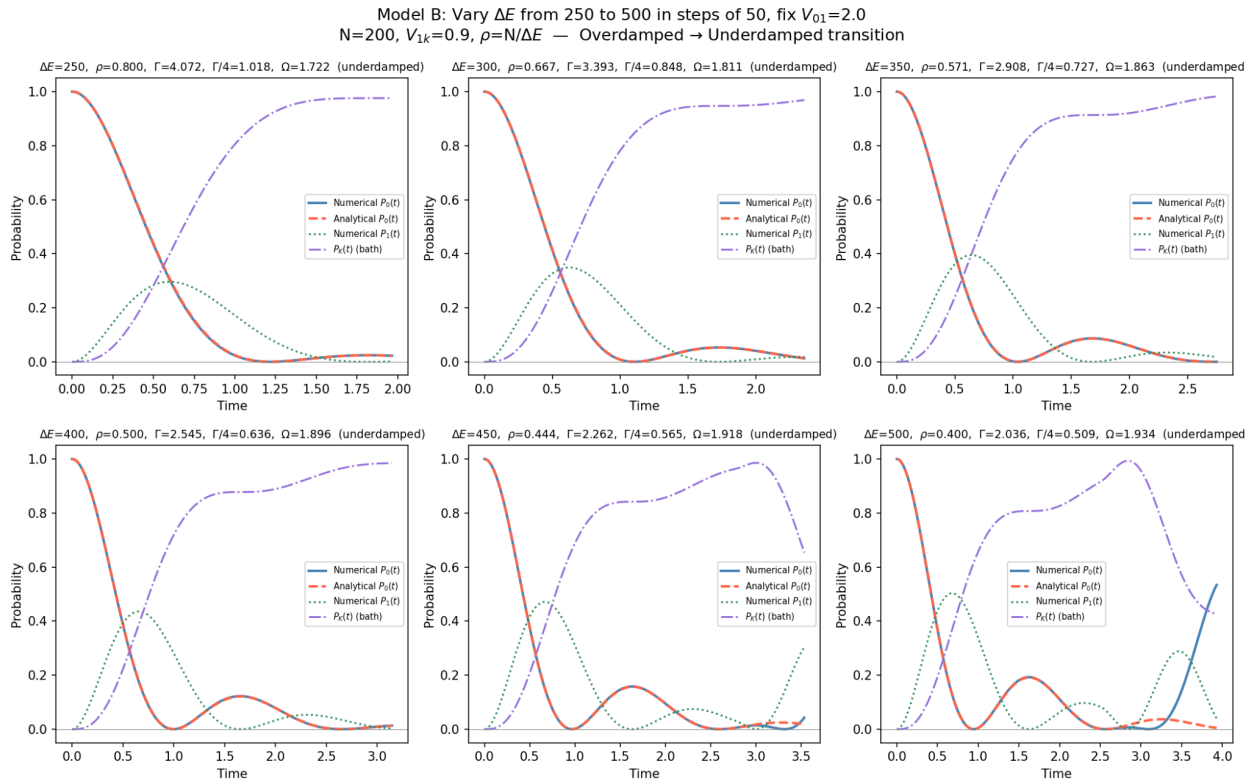


Discussion

For the case where we vary $V_{1,k}$ and hold V_{01} constant we see the same effect as in the previous figure. We move from state $|0\rangle$ to state $|1\rangle$ and then have irreversible decay into $|k\rangle$, and if $\Gamma/4 > V_{01}$ the coupling between $|1\rangle$ and $|0\rangle$ is sufficiently weak to inhibit oscillations, and if $\Gamma/4 < V_{01}$ the coupling between $|1\rangle$ and $|0\rangle$ is sufficiently strong to enable oscillations.

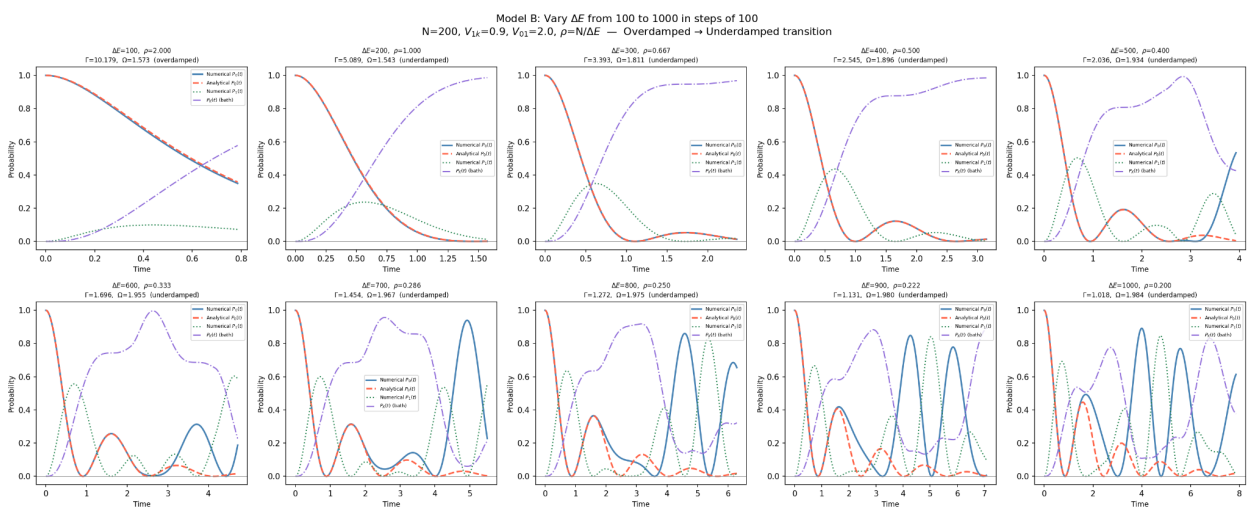
We will also examine how varying ΔE affects our results for System B.

Figure B.3 $N = 200$, $V_{01} = 2$, ΔE varied



Now if we extend ΔE to very large values such as $\Delta E = 100$ we see the model does in fact break from the analytical result as we would expect.

Figure B.4 $N = 200$, $V_{01} = 2$, ΔE varied



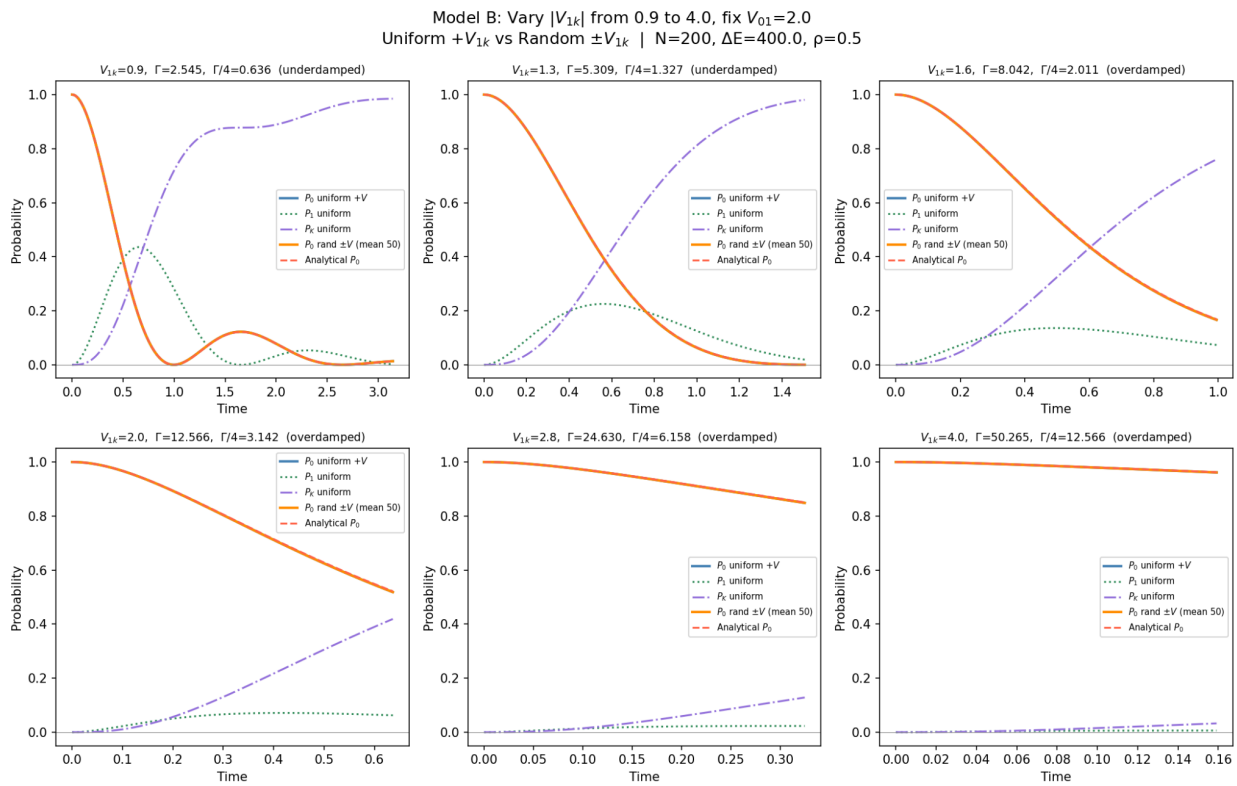
Discussion

We see the same results as we saw before: as ΔE increases for a constant N the results should not change in theory as this is modeling the same system but with varied accuracy

in how well we approximate the infinite continuum of states. However, just like before we reach a point at which the value of ΔE is sufficiently large and sufficiently small to break the model. If ΔE is too large at constant N, then ε becomes too large, and the resolution not being near an infinitely dense continuum means we deviate from the results based on this approximation, i.e. we do not get perfect exponential decay. Then on the other end if ΔE is too small we no longer have the approximation that says we have states going to $\pm\infty$.

We now look at the case where we take a constant $|V_{1,k}|$ but each V is randomly assigned a $+$ or $-$ sign.

Figure B.5 $\Delta E = 400$, $N = 200$, $V_{01} = 0.9$, $\pm|V_{1,k}|$ varied



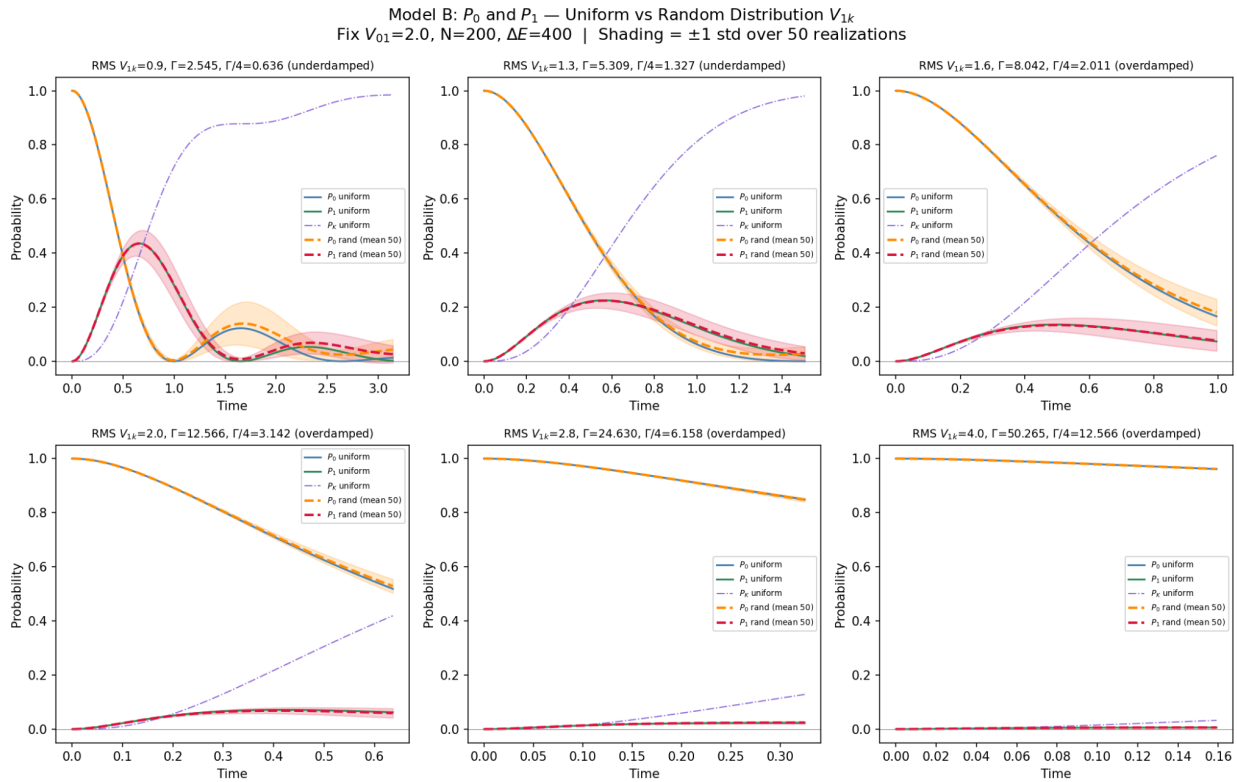
We see these results match those in which V_{01} is constant and exclusively positive.

Discussion

This matches the result we already discussed, namely that changing the sign of various V's will not affect the final results as we have the Fermi Golden Rule which states $\Gamma = 2\pi\rho V^2$; in this we see V is squared and that any sign change cannot affect the final Γ.

We now look at the case where we vary the $V_{1,k}$ values such that $\langle V \rangle = 0$, $\langle V^2 \rangle^{1/2} \gg \varepsilon$, but for System B.

Figure B.6 $\Delta E = 400$, $N = 200$, $V_{01} = 0.9$, $V_{1,k}$ varied on a distribution

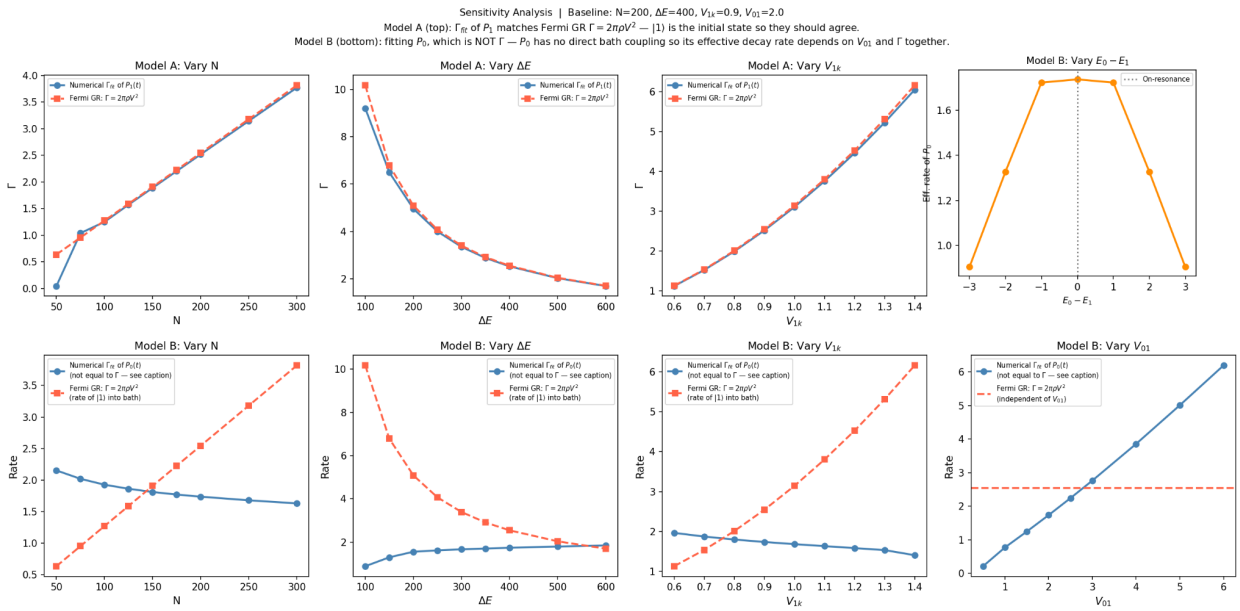


Discussion

These results likewise match those obtained for System A in which we considered a Hamiltonian in which the V's correspond to the values of a distribution. Again, sometimes coupling is stronger and decay occurs more rapidly and other times it is weaker and occurs more slowly, but if we average them we obtain the standard result, with increasingly fast or increasingly slow decay being more and more unlikely the more we deviate from the original system.

We are also to examine the sensitivity of both Systems A and B for E_0, E_1, V_{01}, V_{1k}, N, ΔE. We will do so by varying one variable at a time, as a function of Γ, so that we can directly compare the sensitivity of all of them (decay from $|1\rangle$ to $|k\rangle$). Additionally we compare how the decay rates change as a function of raising or lowering E_1 relative to E_0.

Figure B.7 Γ vs E_0, E_1, V_{01}, V_{1k}, N, ΔE



Discussion

Here I examined Γ as a function of the variables E_0, E_1, V_{01}, V_{1k}, N, ΔE, considering the cases where all but one of the variables is held constant at a time. This method provides a good way of comparing the effects on the decay as Γ is both fundamental to the decay behavior and is connected to all of these variables.

For N we see linear behavior where the strength of the decay increases with greater N, which makes obvious sense as the presence of a greater number of states makes it increasingly more likely that a $|k\rangle$ state and not the $|1\rangle$ state is populated. For increasing ΔE we see decreasing decay strength which is convergent to a value. This makes sense as, since we are holding N constant, increasing ΔE increases ε making our states more spread out/less dense and thus making the individual states further away and more weakly coupled. The shape of this curve is $1/x$ which matches the fact that $\rho = 1/\varepsilon$ with ε increasing linearly with respect to ΔE. Note that this matches the Fermi Golden Rule results.

For System B we will not see agreement between the Fermi Golden Rule, which is actually to be expected as the addition of coupling between $|1\rangle$ and $|0\rangle$ will greatly affect the decay rate. What we do see conceptually is that increasing N has the effect of only slightly decreasing the rate; this means that although adding more states would increase interactions (by simply raising the number of interactions) and thus the decay rate, the decay rate is predominantly affected by the coupling V_{01} and thus is not very sensitive to N for System B. Changing ΔE has very little effect on the decay rate from $|0\rangle$ as the transfer between states $|1\rangle$ and $|0\rangle$ is far more rate-limiting than decay into the continuum. This is likewise true for varying the coupling between $|1\rangle$ and $|k\rangle$. For varying the coupling between $|1\rangle$ and $|0\rangle$, however, we see it increases linearly, which makes sense along the lines we have already stated: the decay rate from $|0\rangle$ is dominated by V_{01} and increases as the coupling increases.

The last case we look at is when we take E_0 and E_1 to not be degenerate by raising and lowering E_1. We see that the largest decay rates are seen for $E_0 = E_1$ and that either

raising or lowering E_1 decreases the rate. This is exactly what we would expect as the most interactions should occur between $|1\rangle$ and $|0\rangle$ when they are most similar in energy, which would then consequently lead to the largest transfer rates and, after eventual transfer to the continuum, the largest Γ.

Note: for E_1, we can use Γ_{fit} as a descriptor as we only ever begin in state $|0\rangle$ for System B, as the outline specifies.